Four-symbol reversal detection on a homogeneous six-state cycle

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25 October 2026 Version 0.4

Abstract

We study a stationary continuous-time nearest-neighbor walk on a homogeneous six-cycle, observed at equal time intervals through the binary labeling 001011. We give an exact formula for the probability difference between the consecutive words 0011 and 1100, and prove that its remaining polynomial factor is strictly positive. Thus this one length-four pair detects every nonresonant time-reversal asymmetry in the specified model. At the resonant intervals the entire observation process is reversible. Length four is sharp because stationary binary words of length at most three are reversal symmetric. Broader reconstruction implications are established prior context; historical novelty of this explicit refinement remains inconclusive.

Unrefereed, AI-assisted research preprint. The mathematical body follows the audited v0.4 note. A received AI review reports independent proof derivation; its reflection condition was subsequently marked PASS, and the remaining FTY source condition was checked author-side against the original appendix. This is not human peer review. Literature coverage is partial; no priority or first-discovery claim is made. The final Howard and Matzinger–Rolles full-text comparisons remain incomplete.

1 Model and statement

Let $X(t)$ be a continuous-time nearest-neighbor Markov chain on $\mathbb{Z}/6\mathbb{Z}$, with clockwise rate $a > 0$ and counterclockwise rate $b > 0$ at every vertex. The initial distribution is uniform. Set $f = (0, 0, 1, 0, 1, 1)$ in clockwise vertex order and $Y_k = f(X(k\tau))$, where $\tau > 0$. A word probability $p(w)$ refers to the stationary law of consecutive Y's, not the reliability of a single observed word.

Put

$$r = e^{-(a+b)\tau/2}, \qquad \theta = \frac{\sqrt{3}}{2}(a-b)\tau, \qquad c = \cos\theta.$$

Candidate theorem. Define

$$G(r, c) = 1 - 2r^2 + 4r^6 + r^8 + 4r^2(1-r^2)c^2 - 2r^3(1+r^4)c.$$

Then

$$p(0011) - p(1100) = -\frac{\sqrt{3}}{54} r^3 \sin\theta \, G(r, \cos\theta),$$

and $G(r, c) > 0$ for $0 < r < 1$ and $-1 \le c \le 1$. Consequently:

1. If $\sin\theta$ is nonzero, this one pair of length-four words witnesses time-reversal asymmetry. No length-one, two, or three consecutive words can witness such asymmetry for any stationary binary process.
2. If $\sin\theta = 0$, the entire sampled hidden chain, and therefore its binary observation process, is time-reversal symmetric at every word length.

For $a \neq b$ the blind spacings are exactly

$$\tau = \frac{2\pi m}{\sqrt{3}\,|a-b|}, \qquad m = 1, 2, 3, \ldots.$$

For $a = b$ all spacings are reversible.

The candidate contribution is the explicit **length-four sufficiency** and its positive prefactor. The all-length dichotomy also follows from established scenery reconstruction, as explained in Section 6; it is not an independent novelty claim.

2 Transition kernel and the four-symbol identity

For the row-generator convention,

$$(Qh)(i) = a(h(i+1) - h(i)) + b(h(i-1) - h(i)).$$

The characters $\exp(2\pi \mathrm{i} ki/6)$ have eigenvalues

$$\mu_k = (a+b)(\cos(2\pi k/6) - 1) + \mathrm{i}(a-b)\sin(2\pi k/6).$$

Thus the six eigenvalues of $P = \exp(\tau Q)$, in character order, are

$$1,\ r(c+\mathrm{i}s),\ r^3(c+\mathrm{i}s),\ r^4,\ r^3(c-\mathrm{i}s),\ r(c-\mathrm{i}s), \qquad s = \sin\theta.$$

Let $K_j = P(0,j)$ and $d = \sqrt{3}$. Fourier inversion gives

$$\begin{aligned}
6K_0 &= 1 + r^4 + 2rc + 2r^3c,\\
6K_1 &= 1 - r^4 + rc + drs - r^3c + dr^3s,\\
6K_2 &= 1 + r^4 - rc + drs - r^3c - dr^3s,\\
6K_3 &= 1 - r^4 - 2rc + 2r^3c,\\
6K_4 &= 1 + r^4 - rc - drs - r^3c + dr^3s,\\
6K_5 &= 1 - r^4 + rc - drs - r^3c - dr^3s.
\end{aligned}$$

For $D_0 = \mathrm{diag}(1-f)$ and $D_1 = \mathrm{diag}(f)$,

$$p(w) = \frac{1}{6}\mathbf{1}^T D_{w_0} P D_{w_1} P \cdots P D_{w_{\ell-1}} \mathbf{1}.$$

Substitution for the two length-four words and use of $c^2 + s^2 = 1$ gives the displayed probability difference. This step can be checked by direct finite expansion: exact_algebra.py enumerates microscopic paths, works over $\mathbb{Q}[r,c,s,d]/(d^2-3)$, and reduces the difference modulo c^2+s^2-1. The coefficient remainder is identically zero. No numerical approximation enters this identity.

3 Strict positivity of G

View G as a quadratic in c. Its leading coefficient is $4r^2(1-r^2) > 0$ and its vertex is

$$c_* = \frac{r(1+r^4)}{4(1-r^2)}.$$

First suppose $r^2 \le 3/4$ and put $t = r^2$. The unrestricted quadratic minimum is

$$G(r, c_*) = \frac{q(t)}{4(1-t)},$$

$$q(t) = 4 - 12t + 7t^2 + 16t^3 - 14t^4 - 4t^5 - t^6.$$

On $0 \le t \le 3/4$ set $u = 4t/3$. The degree-six Bernstein expansion is

$$q(3u/4) = \sum_{j=0}^{6} B_j \binom{6}{j} u^j (1-u)^{6-j},$$

with

$$(B_0, \ldots, B_6) = \left(4, \frac{5}{2}, \frac{101}{80}, \frac{5}{8}, \frac{403}{640}, \frac{443}{512}, \frac{535}{4096}\right).$$

All seven coefficients are strictly positive and the Bernstein basis sums to one. Hence $q(t) > 0$ throughout this interval, proving $G > 0$ there for all real c.

Now suppose $r^2 \ge 3/4$. The function

$$r(1+r^4) - 4(1-r^2)$$

is strictly increasing for $r > 0$, and at $r = \sqrt{3}/2$ has value $25\sqrt{3}/32 - 1 > 0$. Thus $c_* > 1$. On $-1 \le c \le 1$, G decreases as c increases, so it suffices to consider $c = 1$. Exact factorization gives

$$G(r, 1) = (1-r)^2(r^6 + 3r^4 + 6r^3 + 5r^2 + 2r + 1) > 0 \quad (0 < r < 1).$$

This completes the positivity proof. The coefficient identity, vertex minimum polynomial, and endpoint factorization are checked over rationals by verify.py.

4 Sharp word length and the resonant case

For any stationary binary process, the length-two identity $p(01) = p(10)$ follows from equality of one-site marginals. At length three, stationarity of the word 00 gives $p(001) = p(100)$, and stationarity of 11 gives $p(011) = p(110)$. All other length-three words are palindromes. These facts prove the lower bound four without any Markov assumption on Y. This lower bound is established prior art [1].

If $\sin\theta = 0$, all $K_j = K_{-j}$, so the sampled chain has symmetric P. For every hidden path $(i_0, \ldots, i_\ell)$, uniform stationarity and $P = P^T$ make its probability equal to that of its reversed path. Summing over hidden paths with any specified binary labels proves observed reversal symmetry at every length. If $\sin\theta \neq 0$, Section 3 makes the displayed four-word difference nonzero, establishing the converse.

5 Labeling and interpretation limits

A binary labeling f is reflection-symmetric when $f(i) = f(k - i)$ for some k. For uniform stationarity, the reversed hidden transition matrix is $P^T_{a,b} = P_{b,a}$, because $Q^T_{a,b} = Q_{b,a}$; hence the time-reversed observation law at rates (a,b) is the forward observation law at rates (b,a). The reflection $\sigma(i) = k - i$ preserves the uniform initial law, interchanges the two rates, and preserves the observed label whenever $f(i) = f(\sigma(i))$. Thus in this case the forward laws at (a,b) and (b,a) agree, proving reversibility of the observations at every τ, irrespective of drift.

Among the 64 labelings on six vertices, 52 have such a reflection and 12 do not. The latter form the two rotational orbits represented by 001011 and 001101. The second is a rotation of the reversed first labeling. Rotation preserves word probabilities, and reflection interchanges a,b. Thus the theorem covers every chiral labeling on the six-cycle, with a possible sign reversal of the witness.

All binary labelings on cycles of lengths 3,4,5 have a reflection. These small counts are exhaustively verified. The minimal size six is only about **homogeneous nearest-neighbor cycles with deterministic binary vertex labels**. It is not a minimal-state claim about general hidden Markov models.

This note addresses stationary statistical time reversal of sampled paths. It does not say that thermodynamic nonequilibrium is impossible to infer from other structural information. In particular, at odd resonances the sampled matrix has negative eigenvalues, obstructing a reversible continuous-time embedding. At even resonances it equals the sampled propagator of the symmetric ring with rates $a' = b' = (a + b)/2$.

Nor does a length-four witness mean that four recorded bits suffice for a reliable empirical decision. The probabilities need repeated estimation. Signals may be extremely small near resonances or at large τ. No finite-sample detection guarantee or dissipation estimator is claimed.

6 Established implications and remaining novelty question

For Finucane–Tamuz–Yaari [2], the source fixed here is the author-hosted, 17-page manuscript dated 30 March 2014, not an unspecified arXiv version. Definition 1.1 (printed p.2) uses shifts only. Theorem 1.2 and footnote 3 (printed p.3) distinguish this convention from Matzinger–Lember's shift-and-reflection equivalence. The explicit shift-only statement we use is Theorem Appendix A.1 (printed p.14), with its proof completed in Appendix A.3 (printed p.16). It requires the Fourier coefficients to be pairwise distinct **as complex numbers**. A reflection that is not a shift can therefore be distinguished. The original Matzinger–Lember theorem is not silently substituted for this shift-only statement.

Here $0 < r < 1$. The four groups

$$\{1\}, \quad \{re^{i\theta}, re^{-i\theta}\}, \quad \{r^3e^{i\theta}, r^3e^{-i\theta}\}, \quad \{r^4\}$$

have distinct moduli $1, r, r^3, r^4$. Within either conjugate pair, the two complex numbers differ exactly when $\sin\theta \neq 0$. Thus all six Fourier coefficients are pairwise distinct as complex numbers in the nonresonant case; their six absolute values are not pairwise distinct. Applying that theorem distinguishes the chiral labeling from its reflection. At $\sin\theta = 0$ the direct symmetry argument applies. The all-length dichotomy is consequently a known consequence, whereas the explicit one-pair, length-four formula is the candidate refinement.

Periodic-sampling ambiguity itself is also known; Bauer–Seifert–van der Meer [3], Section 3.2.2, give different equilibrium/nonequilibrium generators with one sampled propagator. The general

three-word obstruction is [1]. The remaining novelty question is narrowly whether the explicit sharp four-consecutive-symbol result on this minimal chiral homogeneous ring has already been proved, or follows immediately from another known formula. See PRIOR_ART.md for the recovered Matzinger–Lember manuscript and its version/access limits. The scoped search has not settled historical novelty. REFERENCES are PARTIAL and NOVELTY is INCONCLUSIVE. An independent AI proof-review report has been received within the scope above; human peer review has not been performed. The direct proof in Sections 2-5 does not depend on FTY. Further source-access details are in PRIOR_COMPARISON.md.

References

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[3] M. T. Bauer, U. Seifert and J. van der Meer (2025), *Stroboscopic measurements in Markov networks: exact generator reconstruction vs. thermodynamic inference*. J. Phys. A 58, 125001. https://arxiv.org/html/2412.15642v2 https://doi.org/10.1088/1751-8121/adbc52

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